

YEKATERINO-NIKOL'SKOYE, Russian Federation

WMO: 317070

Lat: 47.7447N

Lon: 130.9703E

Elev: 237

StdP: 14.57

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-16.6	-25.9	1.3	-19.1	-22.5	1.6	-14.9	35.9	3.4	27.6	-2.5	2.0	320	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	86.2	72.0	83.6	70.9	81.1	69.7	75.9	81.8	73.8	79.7	72.0	77.8	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.2	128.6	79.7	72.0	119.5	77.2	70.1	111.6	75.6	39.7	82.1	37.7	80.0	36.1	78.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	17.3	14.1	DB	-25.9	91.5	5.1	3.3	-29.6	93.9	-32.6	95.8	-35.5	97.7	-39.2	100.0	(4)
(5)			WB	-25.3	78.3	5.0	2.9	-28.9	80.4	-31.8	82.1	-34.6	83.8	-38.3	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	-1.5	6.4	22.3	41.5	55.9	65.5	71.6	68.6	57.3	40.9	19.1	1.0
	DBStd	27.02	7.40	8.16	10.29	7.40	7.02	6.10	4.20	4.81	6.38	7.70	11.19	7.87
	HDD50	6724	1595	1220	859	271	23	0	0	0	13	296	928	1519
	HDD65	10462	2060	1640	1324	706	293	66	5	19	239	748	1378	1984
	CDD50	2179	0	0	0	16	206	466	670	576	233	12	0	0
	CDD65	442	0	0	0	0	11	82	210	130	9	0	0	0
	CDH74	3190	0	0	0	13	218	715	1398	765	80	1	0	0
	CDH80	775	0	0	0	1	43	222	367	133	9	0	0	0
Wind	WSAvg	4.3	5.4	4.4	4.9	5.1	4.9	4.0	3.3	3.1	3.4	4.1	4.5	4.9
Precipitation	PrecAvg	24.6	0.2	0.2	0.4	0.9	2.2	3.9	5.4	6.1	3.1	1.2	0.7	0.4
	PrecMax	35.4	0.5	0.9	1.2	6.4	5.5	9.5	12.3	13.1	8.1	4.7	2.4	1.1
	PrecMin	13.0	0.0	0.0	0.0	0.0	0.1	0.4	0.8	0.7	0.4	0.0	0.0	0.0
	PrecStd	6.1	0.2	0.2	0.2	1.1	1.2	2.0	2.6	3.2	2.1	1.0	0.6	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	35.5	53.8	75.4	84.7	91.5	89.6	86.4	80.5	69.3	50.0	25.6
		MCWB	18.0	29.4	42.3	57.1	63.4	70.3	74.3	75.5	68.2	54.0	42.1	22.9
	2%	DB	15.9	29.1	47.2	66.6	79.8	85.6	86.6	83.5	75.8	63.0	42.7	19.9
		MCWB	13.6	24.4	38.5	51.0	60.3	69.4	73.0	73.1	64.2	50.7	36.5	17.6
	5%	DB	12.6	24.2	42.3	60.9	75.4	81.5	84.1	81.0	72.6	58.3	37.6	15.9
		MCWB	10.6	20.9	34.6	47.4	58.5	67.4	72.3	71.4	62.4	47.8	32.6	14.0
	10%	DB	9.6	19.7	37.6	56.1	70.9	77.5	81.7	78.4	69.3	53.6	33.7	12.8
		MCWB	7.9	16.9	31.6	44.7	56.3	65.8	71.4	69.8	60.5	44.9	30.1	11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	29.4	43.2	58.8	66.7	74.7	79.1	78.5	70.5	56.9	43.3	23.5
		MCDB	20.8	34.9	51.3	74.1	79.4	85.2	85.3	84.2	76.2	66.5	49.6	25.3
	2%	WB	13.7	24.8	39.1	52.2	63.4	71.8	76.4	75.7	67.0	52.5	36.9	17.9
		MCDB	15.7	28.6	46.7	64.5	74.7	80.9	82.5	80.3	72.2	60.5	41.6	19.6
	5%	WB	10.8	21.0	34.9	48.7	60.4	69.4	74.5	73.6	64.5	48.7	33.3	14.1
		MCDB	12.6	24.3	41.0	59.4	71.9	78.5	80.7	78.8	70.1	56.3	36.6	15.6
	10%	WB	7.9	17.1	32.2	45.6	57.8	67.3	72.8	71.5	62.1	45.6	30.7	11.2
		MCDB	9.5	19.6	37.5	54.1	68.8	75.3	78.5	76.2	67.5	52.6	34.0	12.7
Mean Daily Temperature Range	5% DB	MDBR	17.1	20.5	20.0	20.7	20.8	17.0	15.6	16.2	19.9	19.3	15.6	15.4
		MCDBR	19.2	23.0	23.4	31.0	29.9	25.4	20.8	20.2	24.8	27.3	20.2	17.1
		MCWBR	17.1	19.2	15.5	17.9	13.8	10.9	9.5	10.3	13.4	15.4	14.3	15.1
	5% WB	MCDBR	19.3	22.9	22.1	27.9	26.1	21.6	17.1	16.5	20.1	25.8	18.6	16.5
		MCWBR	17.3	19.0	15.0	17.1	13.7	10.9	10.2	10.2	12.4	15.9	13.6	14.7
Clear-Sky Solar Irradiance	taub	0.296	0.330	0.414	0.540	0.581	0.542	0.516	0.465	0.426	0.435	0.357	0.305	
	taud	2.145	2.041	1.876	1.730	1.716	1.905	2.022	2.153	2.161	1.985	2.094	2.117	
	Ebn at Noon	239	254	247	226	224	235	239	247	244	212	208	216	
	Edh at Noon	27	38	53	68	71	60	52	44	40	40	29	25	
All-Sky Solar Radiation	RadAvg	619	1014	1423	1516	1672	1744	1645	1521	1268	865	574	477	
	RadStd	41	31	110	159	152	228	175	100	122	63	42	31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page